

1) EXPLAIN THE ORACLE WITH CLOUD COMPUTING.

* The cloud computing services provided by the Oracle Corporation are popular as Oracle Cloud.

* The Oracle Cloud provides storage services, Servers, networks, etc that are delivered to the customer using a global network consisting of a number of Data centers that are managed by the Oracle Corporation.

* Each cloud is deployed in region wherein each region consists of three availability domains that are fault independent in nature.

* There are independent data centers inside each of these independent data centers inside each of these availability domains.

* Like every cloud service provider (CSP), the service offered by the Oracle Cloud are available to the customers on-demand over the internet and the users are charged only for what they use.

* The platform provided by the cloud is compatible with a large number of open source applications such as Hadoop, Kubernetes etc open Standard such as HTML5, SQL etc.

* The services delivered to the customers are broadly categorized as a Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), and Data as a Service (DaaS).

* Infrastructure as a Service (IaaS)

The infrastructure provided by the Oracle cloud is commonly known as Oracle cloud Infrastructure (OCI) which includes:

- * Storage: The Storage services offered include block volumes, object storage and archive storage.
- * Network: The OCI provides the customers with a network which contains IP addresses, routes, subnets, firewalls.
- * Compute: In order to perform compute services, the company offers virtual machine instances of different capabilities according to requirement of workloads.
- * Database: This helps the customer to deploy Oracle databases in the cloud environment.
- * Load balancing: This feature is used to route the traffic automatically across the availability domains (high) availability and fault tolerance.
- * Governance: The OCI helps its users to deal with auditing as well as Identity and Access Management by providing access management checks, data integrity checks and traceability.
- * Ravello: This feature allows the users to deploy their existing VMware or other workloads on OCI and this is done without performing any modifications to the network, storage, and virtual machines.
- * Fast connect: It is used to provide private connectivity on cloud.

* platform as a Service (PaaS)

The platform services provided by the Oracle Cloud are commonly known as Oracle Cloud Platform (OCP).

* Integration: Using this feature, the OCP allows the customers to integrate the cloud application.

* Security: The OCP provides secure access to its customers using security applications. These applications also monitor the hybrid cloud environment along with addressing the other IT requirements.

* Application Development: In order to develop the applications, the OCP provides its customers with an application development platform that is an open standard.

* Business Analytics: This platform provided by the company can be used to analyze the data.

* Management: This platform is based on the machine learning and Big Data operations which is used to provide monitoring, management and Analytics.

* Content and Experience: This platform is used to provide the customers with content collaborations and website and workflow management. This is an inbuilt feature that comes with the Oracle SaaS Services. These are available under the name such as DIVA cloud, Experiana cloud, etc.

* Software as a Service (SaaS).

The SaaS Applications provided by the cloud are commonly known as Oracle Cloud Applications (OCA) which includes following.

- * SaaS Analytics

- * Enterprise Resource planning (ERP)

- * Enterprise Performance Management (EPM)

- * Internet of Things (IOT)

- * Customer Experience (CX)

- * Blockchain cloud service etc.

* Data as a Software (Daas) :

The platform services provided by the Oracle cloud commonly known as Oracle Data cloud (ODC) this platform is used to Analyze the customer data that is powered by the Oracle ID graph.

- * Drop box

- * Samsung

- * Zoom video Communications

- * Yamaha Corporation

- * Yum! Brands

2) EXPLAIN ABOUT ORACLE AI DATABASE :

* Oracle AI Database is a modern database system developed by Oracle Corporation that integrates artificial intelligence (AI) directly into the database.

* It is designed to automate database management tasks and support AI-driven applications.

* The Oracle AI Database runs mainly on Oracle cloud infrastructure (OCI) and provides intelligent automation, improved performance, and enhanced security for managing large amounts of data.

* Oracle AI Database is an advanced version of the Oracle database that uses AI and machine learning to automatically perform tasks such as

- * Database tuning

- * Performance optimization

- * Security monitoring

- * Query optimization

- * Automatic updates and backups.

* It reduces the need for manual database administration.

* Autonomous Database :

The Database can self-manage, self-repair, and self-secure using AI algorithms. It automatically performs tasks such as patching, backups and scaling.

* AI Vector Search

Oracle AI Database includes vector search technology which helps in building AI applications like

Recommendation Systems, chatbots and Semantic Search engine.

* Automated Performance Tuning:

AI analyzes database workloads and automatically optimizes queries and indexes to improve Performance.

* High Security:

AI continuously monitors database activity and detects threats and suspicious behaviour.

→ components of Oracle AI Database.

- * Data storage layer - stores structured & unstructured data
- * AI/ML engine - performs machine learning operations
- * Query processing engine - Executes database queries
- * Security system - Protects data using encryption and monitoring tools.

→ Advantages:

- * Automatic database management
- * Improved performance and scalability
- * Reduced operational cost
- * Advanced Data Analytics
- * Better security
- * Threat detection

→ Applications:

- * Banking and Financial Services
- * Healthcare Data Analysis

- * E-commerce recommendation systems
- * Fraud detection systems
- * Smart business analytics.

* Oracle AI Database is an intelligent Database platform that combines traditional database management with artificial intelligence capabilities. By automating tasks such as optimization, security and maintenance, it reduces administrative workload and enables organizations to build powerful AI-driven applications more efficiently.